

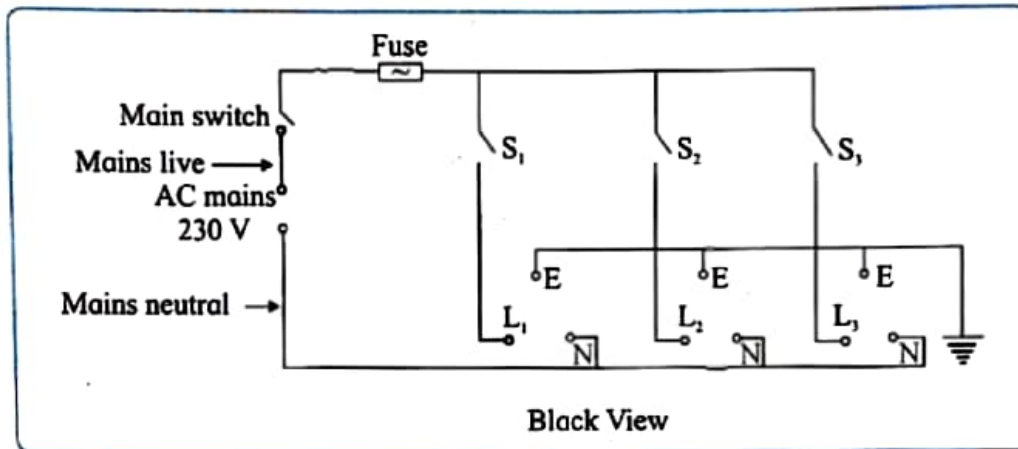
ACTIVITY NO. 6

HOUSEHOLD CIRCUIT

Aim : To design, build and demonstrate a simple house hold electrical circuit .

Apparatus : - ON-Off switches, three pin sockets, flexible connecting wires, bulbs, fuse, power supply etc.

Diagram :



S_1 , S_2 , S_3 - ON-Off switches, L_1 , L_2 , L_3 - Live terminals, N- Neutral terminals, E- Earthing

Procedure :

1. Connect the live wires L_1 , L_2 , L_3 in series with the switches S_1 , S_2 , S_3 respectively.
2. Connect each set of Live wire and switch in parallel combination with each other as shown in circuit diagram.
3. Connect the fuse in series with main switch and A.C. mains.
4. Join all neutral wires together and connect to the neutral terminal of AC mains.
5. Similarly join all earthing wires together and give it to the earthing connection.
6. Check the circuit once again to ensure that household circuit is complete.

Result: Household circuit is verified.

Precautions :

1. Live wire and neutral wire should be well separated.
2. While making connection do not connect the wire of A.C. mains.
3. All connections should be tight.

Questions

1. What is household circuit?

A household circuit is the electrical wiring system used in a house to supply electricity to lights, fans, sockets, and other appliances. It consists of three wires: live wire, neutral wire, and earth wire.

2. How are the circuits wired in houses?

Household circuits are generally wired in parallel. This allows each appliance to operate independently, and if one appliance fails, the others continue to work.

3. What is the importance of earthing in household circuit?

Earthing protects people from electric shock by providing a low-resistance path for leakage current to flow safely into the ground. It also helps protect electrical appliances from damage.

4. Explain the role of fuse in household circuit.

A fuse is a safety device that protects a circuit from excessive current. When the current exceeds the safe limit, the fuse wire melts and breaks the circuit, preventing overheating, fire, and damage to appliances.

Remark and sign of teacher :